

Scenario Catalog

PREPARED BY Energy Hippo implementation team

Each scenario should describe behavior the system must handle. Source data is only useful when the expected downstream result is clear.

V1 Scenarios

Scenario	Purpose	Initial scope	Expected assertions
baseline_monthly_bills	Prove standard bill ingestion and rollups.	24 months across electric, gas, water, sewer, and chilled water.	Bill totals match source model by account, building, site, and customer.
electric_interval_15min	Prove interval ingestion and time-series rollups.	Science Center electric channels through MV90/MDEF and Student Center electric channels through AcquiSuite.	Interval count, usage totals, peak demand, gateway point mappings, and local time handling match expected values.
solar_generation_interval	Prove generation interval ingestion and sign convention.	Helios PV generation/export channels through MV90/MV9.	Generated kWh, exported kWh, REC basis, and net-load reduction match expected values.
chilled_water_hourly	Prove non-electric interval shape.	Science Center chilled-water ton-hours and tons through BACnet present values.	Usage totals, demand, unit conversion, and BACnet point mappings match expected values.
weather_gateway_noaa	Prove NOAA current-observation ingestion.	KSAC temperature, humidity, and wind points.	NOAA station id, weather type mapping, local time, and record counts match expected values.
weather_gateway_aeris	Prove Aeris observation/forecast ingestion.	KSMF observed and forecast temperature points.	Request counts, state cursor, forecast gating, and point bindings match expected values.
spinwave_gateway	Prove Spinwave CSV ingestion.	Science Center wireless temperature and humidity sensors.	Folder/device/channel mappings and interval counts match expected values.
fig_gateway_formats	Prove FIG-family provider file ingestion.	One Cedar Row water alias emitted in FIG, SIEMANSREPORT, MEDIATOR, EATON, XML, FIXEDNETWORKS, and CMEP shapes.	Each format binds to the same source channel and produces equivalent values.
modbus_gateway	Prove Modbus TCP point reads.	Thermal plant chiller kW from a floating-point register.	Register type, address mapping, scaling, and timestamping match expected values.
odbc_gateway	Prove SQL historian query ingestion.	Library electric interval reads through an ODBC profile and query key.	Query token substitution, cursor state, and backfill windows match expected values.

Scenario	Purpose	Initial scope	Expected assertions
fake_gateway	Prove deterministic generated gateway data.	One synthetic 15-minute kWh channel.	Seeded generation, state handling, and optional anomaly/replay shapes match expected values.
handheld_gateway_uploads	Prove handheld upload ingestion.	Neptune water and MVRS gas route uploads.	MetaWorldID binding, usage/register point mapping, constants, and read timestamps match expected values.
hmr_events_publish	Prove handheld event publishing.	MVRS event output file published to the HMR event broker shape.	Event count, idempotency scope, queue metadata, and gateway identity match expected values.
meter_replacement	Prove meter lifecycle handling.	One electric account changes meter IDs mid-stream.	Old and new meters map to the same building without double counting.
estimated_bill_correction	Prove corrected bill behavior.	One natural gas estimated bill followed by a correction.	Corrected bill supersedes or adjusts the estimate according to product rules.
missing_interval_backfill	Prove late-arriving interval data.	A gap in Student Center interval reads is backfilled later.	Gap detection, backfill ingestion, and final totals match expected values.
duplicate_file_import	Prove idempotency.	Re-import one generated provider file.	Second import does not duplicate usage, costs, meters, or bills.
account_number_change	Prove account continuity.	Water account changes account number at renewal.	New account identifier preserves site and building continuity where expected.
provider_rate_change	Prove rate-sensitive calculations.	Electric rate changes at the start of a fiscal year.	Bills after the change use the expected tariff metadata.
negative_bad_units	Prove validation failures are explicit.	One intentionally invalid provider file uses an unsupported unit.	Import fails with the expected validation error and does not partially load data.

Scenario Rules

- Positive scenarios should be valid and repeatable.
- Negative scenarios must be labeled and isolated from normal demo loads.
- Every scenario must define source rows, generated artifacts, and validation assertions.
- Scenario names should be stable because generated artifact names and test results may reference them.